

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1510

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Code of Conduct

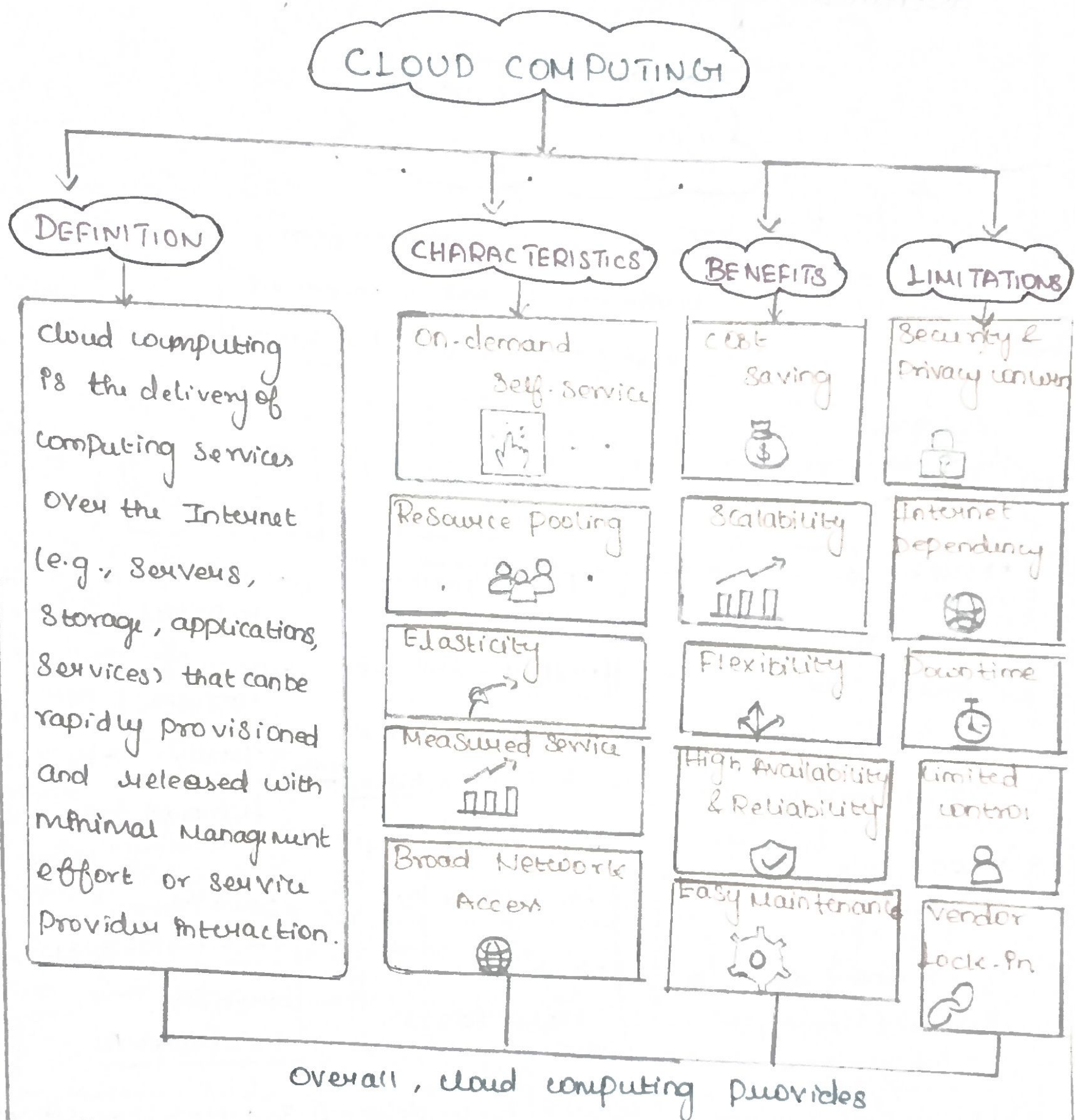
I K. Sai Priya (92421296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sai Priya

Concept Map 1: Cloud computing fundamentals.

Includes Definition, characteristics, benefits and limitations



• Efficient resource utilization | • Business agility | • Innovation enablement while managing trade-offs.

2. Concept map 2: Evolution of cloud computing.
connect hardware evolution, Internet software evolution, distributed systems, and service delivery.

EVOLUTION OF CLOUD COMPUTING

The journey of computing from traditional systems to on-demand cloud services over the Internet

1. Hardware Evolution

- Mainframes
- client-server Architecture
- virtualization (vm, Hypervisor)
- Multi-core, Cluster computing
- Data centers & Resource Pooling

enables

Efficient utilization of hardware resources

2. INTERNET & SOFTWARE EVOLUTION

- Early Internet (connectivity)
- web 1.0 - Static web
- web 2.0 - Dynamic web, APIs, AJAX
- Service-oriented Architecture (SOA)
- Web Services (SOAP, REST)

enables

Software services delivered over the Internet

3. DISTRIBUTED SYSTEM

- Grid computing
- utility computing
- Parallel & Distributed Processing
- Fault Tolerance & High Availability
- Large Scale Data Management

Scalable, reliable and always available systems

4. SERVICE DELIVERY (cloud)

- On-demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service

cloud services IaaS, PaaS, SaaS, XaaS



Outcome: Evolution leads to cloud computing - providing flexible, Scalable and On-demand Services

Concept Map 3: Service models IaaS, PaaS, SaaS, XaaS Map with Provider responsibilities and user responsibilities

CLOUD SERVICE MODELS

cloud service models define the type of service delivered over the cloud and the division of responsibilities between provider and user.



Outcome: Enables efficient, scalable, and cost-effective cloud services through shared responsibilities

Concept Map 1: Virtualization and web services as Enablers of cloud. Show these two concept support cloud deployment and scalability.

